Benchmarking Paired fMRI Perception–Imagery Decoding Under Overlap Scarcity

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Abstract

Paired perception–imagery decoding from fMRI is scientifically important but evaluation-poor. When perception and imagery sessions share only a handful of overlapping stimuli, the question shifts from *which decoder is best?* to *which conclusions are even warranted?* We contribute a frozen benchmark and evaluation surface for this overlap-scarce regime—not a new decoder architecture. The benchmark comprises 94 mixed rows, 5 shared paired stimulus identifiers, and 1 held-out paired evaluation group across four subjects from the Natural Scenes Dataset. We evaluate a fixed comparison ladder of three decoder families: Ridge regression (external low-data reference), a shared-only neural decoder (canonical neural baseline), and shared-private neural variants (exploratory hypothesis family). On this surface, Ridge is the strongest overall baseline (test cosine similarity 0.552, test MSE 0.00117), shared-only is the strongest canonical neural model (0.136, 0.00225), and the best shared-private variant remains below shared-only (0.108, 0.00232). These results yield a deliberately qualified conclusion: the current evidence does not support explicit disentanglement as the preferred model family under severe overlap scarcity. We are not aware of a prior frozen benchmark for paired perception–imagery decoding under explicit overlap constraints, or one that pre-declares evidential roles before evaluation. The contribution is therefore a frozen evaluation surface with pre-declared model roles, a reusable comparison ladder, and a claim-boundary methodology documenting which conclusions the current regime does and does not support. Future studies with larger paired data can rerun the same protocol without changing the evaluation contract.

1 Introduction

The central bottleneck in paired perception–imagery brain decoding is not architectural: it is evaluative. When the number of stimuli shared between perception and imagery sessions is extremely small, standard model comparisons become fragile, held-out evaluation sets shrink to the point of unreliability, and more expressive architectures become easy to overinterpret. This *overlap scarcity* problem is not a local inconvenience in one particular dataset. It is a structural feature of realistic paired neuroimaging settings, where the intersection between externally driven and internally generated responses is filtered through subject availability, stimulus overlap, and split constraints [Pearson, 2019, Horikawa and Kamitani, 2017, Dijkstra et al., 2018].

Despite this, much recent fMRI decoding work has focused on architectural sophistication or reconstruction quality rather than on whether evaluation surfaces actually support the conclusions drawn from them [Takagi and Nishimoto, 2023, Radford et al., 2021]. This creates a particular risk in paired perception–imagery settings, where shared-private or disentangled models carry an appealing narrative—that perception and imagery can be factored into shared content and domain-specific components—but that narrative requires sufficient paired evidence to justify. When paired supervision is scarce, transfer diagnostics become fragile, held-out support is narrow, and expressive models become easy to overinterpret. A conceptually attractive shared-private model may still be empirically unjustified if it is evaluated on too little paired data or against an insufficiently disciplined baseline ladder.

This paper treats evaluation discipline as the primary contribution rather than model novelty. We freeze a benchmark, report the current model ordering honestly, and document the claim boundary that the present paired regime supports. The benchmark is real but severely constrained: 94 mixed rows across four subjects, with only 5 shared paired stimulus identifiers and 1 held-out paired evaluation group. On this surface, Ridge regression is the strongest overall baseline, a shared-only neural decoder is the strongest canonical neural model, and shared-private variants remain exploratory.

We argue that this is a useful contribution to the NeurIPS Evaluations and Datasets track for three reasons. First, no frozen benchmark surface currently exists for paired perception–imagery decoding under overlap scarcity. Researchers who wish to compare models in this regime must build ad hoc evaluations whose data regime, target space, and baseline roles vary between studies—precisely the kind of benchmark fragility that Dehghani et al. [2021] showed can reverse model rankings in other domains. Second, the benchmark's qualified findings directly prevent a concrete failure mode: without a stable reference ladder, future work may attribute neural-model gains to disentanglement rather than to changes in dataset scale, because no fixed comparison point exists. Third, the evaluation protocol is designed for reuse. Its pre-declared evidential roles and fixed target space mean that when larger paired overlap becomes available, the same ladder can be rerun and directly compared to the current ordering without redesigning the evaluation.

Contributions.

- A reproducible ROI-first benchmark for multi-subject paired perception–imagery decoding, with canonical preparation, preflight, training, evaluation, and export workflows.
- A frozen comparison ladder that assigns fixed evidential roles—external reference baseline, canonical neural baseline, exploratory hypothesis family—before interpretation begins.
- An honest evidence report under severe overlap scarcity: Ridge is strongest overall, shared-only is the strongest canonical neural model, and shared-private disentanglement is not yet justified as the preferred family.
- A frozen claim boundary that documents what can and cannot be concluded from the current benchmark, designed to prevent overclaiming and to motivate—but not confirm—a future threshold hypothesis.

Absent this benchmark, the practical risk is concrete: paired perception–imagery decoding would continue without a frozen comparison point. Architectural conclusions would rest on evaluations whose data regime, target space, and baseline roles vary between studies, making it impossible to determine whether a reported improvement reflects genuine progress or a change in evaluation conditions. This benchmark does not solve the data-scarcity problem. It makes the problem legible and auditable, so that future work can build on a stable foundation rather than on shifting sand.

2 Related Work

Visual content decoding from fMRI. Large-scale fMRI datasets such as the Natural Scenes Dataset [Allen et al., 2022], BOLD5000 [Chang et al., 2019], and THINGS-data [Hebart et al., 2023] have enabled rapid progress in visual content decoding. Early deep-learning approaches demonstrated that hierarchical visual features can be recovered from brain activity [Shen et al., 2019], and recent reconstruction systems leverage latent diffusion models and large pretrained vision-language encoders to generate high-fidelity images from fMRI [Takagi and Nishimoto, 2023, Ozcelik and VanRullen, 2023, Scotti et al., 2024]. These systems achieve impressive perceptual quality but are evaluated almost exclusively on perception-only decoding with abundant training data. Tang et al. [2023] extended decoding to continuous language, further raising expectations for what brain decoders can achieve. Yet the harder paired perception–imagery regime—where overlapping stimuli between conditions are scarce—remains comparatively underserved in terms of evaluation discipline.

Perception–imagery transfer and disentanglement. Horikawa and Kamitani [2017] demonstrated that hierarchical visual features can be decoded from both seen and imagined objects, establishing the basic scientific question. Pearson [2019] and Dijkstra et al. [2018] further characterized the neural overlap and divergence between perception and imagery. These studies motivate shared-private or disentangled modeling, but the datasets available for such modeling are typically small, evaluation surfaces have not been frozen in a way that separates architectural conclusions from data-support limitations, and no prior work has pre-declared the evidential roles of competing model families before reporting paired results.

Benchmark methodology. The broader ML community has recognized that benchmark design itself is a source of fragility. Dehghani et al. [2021] showed that relative model rankings can change substantially depending on which evaluation tasks are included, underscoring the importance of fixing the evaluation surface before drawing architectural conclusions. Large-scale holistic evaluations [Liang et al., 2023] have demonstrated the value of standardized, transparent comparison ladders across diverse tasks. Our contribution applies a similar principle to a much smaller but structurally important domain: paired perception–imagery fMRI decoding, where no standardized evaluation surface currently exists.

Multi-subject alignment. Naselaris et al. [2011] emphasized principled encoding and decoding comparisons, and multi-subject alignment methods [Haxby et al., 2011] have shown that representational comparisons require careful normalization. Our benchmark adopts an ROI-first input contract rather than voxel-wise alignment, which is sufficient for the comparison-ladder question addressed here.

3 Benchmark Design

The scientific object in this paper is a paired benchmark under constrained support. We study a regime in which perception and imagery are aligned through a shared stimulus identifier, but only for a very small set of paired items.

Data regime and overlap scarcity. The benchmark is derived from the intersection of two public neuroimaging resources: canonical perception indices from the Natural Scenes Dataset [Allen et al., 2022] and mental imagery response data from the public NSD-Imagery acquisition. After canonical filtering, overlap assembly, and multi-subject intersection, the resulting prepared dataset contains 94 rows across subjects `subj02` (38 rows), `subj03` (18), `subj05` (19), and `subj07` (19), with 5 shared paired stimulus identifiers and 1 held-out paired evaluation group.

This scarcity is not an artifact of poor data engineering. It reflects a genuine ceiling on the overlap between currently accessible perception and imagery sessions after principled filtering. The benchmark's checked-in preflight policy classifies this as below *confirmatory* paper-scale readiness (which requires ≥ 32 paired groups for strong positive claims about disentanglement). We report this threshold transparently and treat the benchmark accordingly: it is appropriate for freezing a comparison ordering and documenting a claim boundary, but not for strong confirmatory statements. The gap between the threshold and the current data is itself an important finding: it quantifies how much more paired support the community needs before shared-private conclusions become defensible.

Fixed target space and input contract. Every model in the ladder predicts the same target representation, `vit_l14_image_768`, so the benchmark compares decoder structure rather than target-space drift. The multi-subject input contract is ROI-first: fMRI features are organized into three ROI groups—early visual, ventral visual, and metacognitive—rather than raw full-fMRI vectors. This is consistent with representational rather than voxel-wise multi-subject alignment [Haxby et al., 2011] and necessary for valid multi-subject batching across subjects with different raw voxel dimensionalities.

Readiness and scope. Overlap scarcity matters for three reasons, each specific to the paired perception–imagery setting rather than to low-data regimes in general. First, it limits what can be inferred from held-out *paired* evaluation: a small number of paired groups makes transfer-like claims and domain-specific diagnostics inherently weak, in a way that single-domain scarcity does not.

Table 1: Frozen comparison ladder on the max-available paired overlap set (94 rows, 5 shared paired IDs, 4 subjects). Bold marks the best value in each metric. Dashes indicate condition-level means absent from the frozen evidence bundle. "Groups" is the number of held-out paired evaluation groups. Evidential roles are pre-declared.

System	Evidential role	Cosine↑	MSE↓	Img. mean	Perc. mean	Groups
Ridge	External low-data reference	**0.55199**	**0.001167**	0.55152	0.55446	1
Shared-only	Canonical neural baseline	0.13596	0.002250	0.13422	0.14527	1
Exploratory shared-private family						
SP, $d_{\mathrm{priv}} = 16$	Best SP variant tested	0.10784	0.002323	—	—	1
SP, $d_{\mathrm{priv}} = 8$	Recovery variant	0.09595	0.002354	—	—	1
SP default	Hypothesis-family baseline	0.06927	0.002424	0.07151	0.05735	1
SP, no domain head	Diagnostic control	0.05907	0.002450	—	—	1

Second, it changes the burden placed on model capacity asymmetrically: when paired supervision is sparse, shared-private factorizations pay a capacity penalty that shared-only or linear models do not, because the private branches must be justified by paired evidence that may not exist. Third, it increases the importance of comparison discipline: if data preparation, target spaces, or model roles drift between baselines, the benchmark stops answering the intended scientific question about shared versus private structure.

4 Comparison Ladder and Evaluation Protocol

The benchmark assigns fixed evidential roles to each model family *before* interpretation. This pre-declaration is a deliberate methodological choice: when a comparison surface is small, post-hoc role reassignment (e.g., reclassifying a failed primary model as a "diagnostic control") can disguise negative results as designed ablations. By fixing roles in advance, the benchmark preserves the scientific meaning of the ordering regardless of which model wins.

Model roles. *Ridge regression* is the external low-data reference baseline. Simple linear models remain strong comparators in fMRI decoding [Naselaris et al., 2011], and Ridge serves as an honest floor: if an expressive neural model cannot beat Ridge under scarce overlap, the architectural complexity is not yet warranted. *Shared-only* is the canonical neural baseline. It uses the same ROI-first input and target space as the shared-private family but removes explicit private branches and domain classification, providing the simplest neural comparator. *Shared-private variants* are the exploratory hypothesis family. They carry the heaviest burden of proof: a more expressive factorization requires evidence that it helps rather than simply fits the task narrative. Three shared-private variants are evaluated: the default model, a reduced-capacity variant with `private_dim=16`, and a `private_dim=8` recovery variant. A no-domain-head ablation (shared-private with domain classification disabled) is included as a diagnostic control.

Hyperparameters and metrics. All neural runs use the same checked-in hyperparameter family: batch size 8, 5 training epochs, learning rate 10^{-3}, weight decay 10^{-4}, and seed 0. Primary metrics are content cosine similarity and content MSE on the held-out evaluation set. Condition-level imagery and perception means are reported where present in the frozen evidence bundle. Transfer summaries and domain accuracy remain secondary diagnostics only: with 1 held-out paired group they are too weak to sustain a headline claim.

5 Results

The main result is a benchmark ordering, not a model novelty claim. Table 1 reports the frozen ladder. We organize the findings into three tiers by evidential strength.

Strongly supported findings. First, the ROI-first benchmark is operational on real paired data and produces a stable, reproducible comparison ladder. Second, within the present low-overlap regime, simpler models dominate. Ridge is the strongest overall baseline (test cosine 0.55199, test MSE

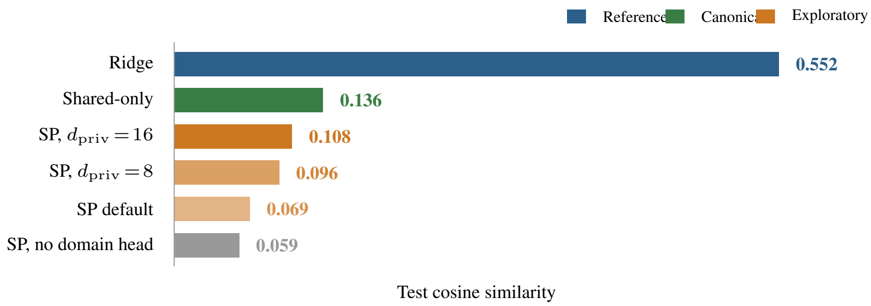

Figure 1: Frozen benchmark ladder on the max-available paired overlap set. Bar length is proportional to test cosine similarity. Ridge dominates all neural models; shared-only is the strongest canonical neural baseline; all shared-private (SP) variants remain below shared-only. Color encodes the system's pre-declared evidential role.

0.001167), and shared-only is the strongest canonical neural model (0.13596, 0.002250). The gap between Ridge and all neural models is large, indicating that linear regression remains a formidable comparator when paired training data is severely limited.

Diagnostic findings. The shared-private story is more limited but still informative. Reducing private capacity helps within that family: both `private_dim=16` (0.10784) and `private_dim=8` (0.09595) improve over the default shared-private model (0.06927). However, none overtakes shared-only. Disabling the domain head alone does not rescue shared-private performance (0.05907), suggesting that the gap is not attributable to a single broken auxiliary component. Instead, the pattern is consistent with a scarce-overlap regime that penalizes unnecessary capacity, regardless of where that capacity sits.

Unsupported conclusions. The current results do not show that shared-private disentanglement improves content decoding. They do not show that shared-private structure improves transfer generalization. They do not show that private branches already isolate neuroscientifically meaningful perception-private or imagery-private factors. And they do not confirm a threshold at which disentanglement begins to help.

Takeaway. The benchmark supports a disciplined ordering, not a crossover claim. Ridge remains the strongest reference point, shared-only is the correct current neural baseline, and shared-private remains exploratory. This is not a failed method paper reframed as a benchmark: the benchmark was designed to determine whether disentanglement is justified under scarce paired overlap, and the answer is that it is not yet justified. That qualified conclusion is the paper's scientific contribution.

6 Discussion

What the benchmark establishes. The frozen benchmark establishes three things. First, a reproducible evaluation surface exists for paired perception–imagery decoding under overlap scarcity: the preparation, training, evaluation, and comparison workflows are operational and auditable. Second, the current model ordering is clear and stable: Ridge $>$ shared-only $>$ shared-private. Third, several tempting architectural conclusions are ruled out: explicit disentanglement does not improve content decoding in this regime, shared-private structure does not improve transfer generalization, and private branches do not yet capture neuroscientifically interpretable structure.

Why Ridge dominates. The $4\times$ gap between Ridge (0.552) and the best neural model (0.136) is large but not unexpected. Linear ridge regression is a consistently strong baseline in fMRI decoding with limited training data [Naselaris et al., 2011], and the effect is amplified in our setting for two structural reasons. First, the training set contains only 56 rows after splits, which is well below the regime where over-parameterized neural models outperform regularized linear methods on typical fMRI feature spaces [Horikawa and Kamitani, 2017]. Second, Ridge operates on an ROI-feature space with only 39 dimensions, where the bias–variance trade-off strongly favors a high-bias estimator.

Neural models must learn shared (and, optionally, private) representations from the same 56 rows; the additional capacity is not warranted by the available supervision. The gap is therefore *a finding of the benchmark*, not a limitation of the neural models: it quantifies how much low-overlap regimes favor simplicity, and it establishes the baseline that future, larger paired datasets must beat before architectural sophistication can be justified.

What is suggestive but not confirmed. The pattern of improvement within the shared-private family as private capacity decreases is consistent with the hypothesis that disentanglement may begin to help only when paired support is large enough to sustain the additional capacity. This motivates a *threshold hypothesis*—the idea that there exists some overlap scale at which shared-private models cross above shared-only—but the present evidence only motivates that hypothesis; it does not confirm it.

Why negative findings matter for E&D. Overclaiming is a persistent risk in brain decoding, where strong narratives about what architectures "should" do can outrun the evaluation data. Consider the concrete failure mode this benchmark prevents: a future study might report that a shared-private model improves paired decoding, but if that study uses a different data regime, a different target space, or a different baseline ladder, there is no way to tell whether the gain came from the architecture or from a change in evaluation conditions. A frozen benchmark with an honest ordering makes this kind of confound detectable: the same ladder can be rerun under the new conditions, and any performance change can be attributed to the data or architecture rather than to benchmark drift.

Why this community, why now. The NeurIPS Evaluations and Datasets track is the right venue because the core problem is evaluation methodology under data-support mismatch, not model novelty. Dehghani et al. [2021] showed that benchmark design choices can reverse model rankings; this work applies the same lesson to a setting where no standardized benchmark exists at all. The question of when benchmark evidence is sufficient to support architectural conclusions is a general ML concern, but it is especially acute in brain decoding, where paired supervision is structurally scarce and strong narratives about shared-private factorization can outrun the data. This benchmark is timely because recent reconstruction-focused work [Takagi and Nishimoto, 2023, Ozcelik and VanRullen, 2023, Scotti et al., 2024] has raised expectations for what brain decoders can achieve, increasing the risk that paired perception–imagery claims will be made without adequate evaluation surfaces.

Reproducibility as contribution. The benchmark is tied to checked-in canonical configs, a fixed target space, explicit workflow entry points, and named artifact roots for preparation, training, evaluation, and export. This matters because a negative or qualified result is only informative if reviewers can see that the comparison surface was disciplined rather than ad hoc. The paper's main claims are grounded in four artifact-bearing surfaces: the prepared mixed overlap dataset, the shared target cache, the baseline/model artifact roots, and the preflight status that records why the benchmark remains below paper-scale support. Appendix A provides the config-command-artifact contract for the main stages.

7 Threats to Validity

We organize the main threats by source and then explain why they are compatible with a valid E&D contribution.

Overlap scarcity. The benchmark contains only 94 rows and 5 shared paired stimulus identifiers. This is the primary limitation. It means the results are adequate for freezing a benchmark ordering and ruling out some stronger claims, but not for establishing a general law about when disentanglement should help. The preflight policy classifies this regime as below paper-scale readiness (which requires ≥ 32 paired groups). We treat this limitation as part of the contribution rather than as a weakness to hide: the overlap scarcity problem itself is what the benchmark makes legible.

Held-out support weakness. With only 1 held-out paired evaluation group, transfer diagnostics and domain accuracy are too fragile to support headline conclusions. We report them as secondary diagnostics only and avoid basing claims on them.

Table 2: Claim boundary for the frozen benchmark. Each row classifies a potential conclusion and states what additional evidence would be needed to upgrade it. The paper is strongest as an evaluation and reproducibility contribution, not as a positive disentanglement result.

Status	Conclusion	Evidence needed to upgrade
Supported	The benchmark provides a reproducible ROI-first evaluation surface for multi-subject paired perception–imagery decoding.	None; operational claim is validated.
Supported	Ridge is the strongest model on the current low-overlap benchmark.	None; frozen ordering on the current surface.
Supported	Shared-only is the strongest canonical neural baseline.	None; frozen ordering on the current surface.
Partial	Low-overlap regimes favor simpler decoders over explicit disentanglement.	Repeated evidence on larger paired datasets or controlled overlap sweeps.
Partial	Private-capacity scaling matters in low-data regimes.	More capacity settings and reruns on larger data.
Not justified	Shared-private disentanglement improves content decoding.	Materially larger paired benchmark where SP beats shared-only.
Not justified	The threshold hypothesis is confirmed.	Overlap-expansion study showing a transition in model ordering.
Not justified	Private latents are neuroscientifically meaningful.	Larger paired support, stable ROI analyses, repeated evidence.

Limited statistical breadth. The frozen results in Table 1 come from a single run per model configuration. A supplementary seed-stability check (Appendix D) re-runs the two primary neural models with three global random seeds and confirms the ordering. A bootstrap analysis over the 19-row test set (Appendix E) provides 95% confidence intervals: the Ridge–shared-only difference (0.45, CI [0.40, 0.49]) and the shared-only–SP $d_{\text{priv}} = 16$ difference (0.081, CI [0.074, 0.087]) both exclude zero, indicating that the ordering is unlikely to be a sampling artifact. Three seeds are not a formal significance test, and 19 test rows limit statistical power, but together they move the benchmark from a single-run snapshot toward a reproducibly stable ordering.

Dependence on recoverable artifacts. The benchmark relies on the particular subjects, splits, and public-data artifacts that were recoverable from the current NSD and NSD-Imagery releases. Different subject pools or data versions could yield different overlap sets.

Absence of confirmatory threshold evidence. The threshold hypothesis remains motivated but unconfirmed. This paper does not test it; it only identifies the preconditions for testing it.

Why the benchmark is still useful. These limitations do not make the benchmark premature; they make it necessary. The overlap scarcity documented here is not a temporary data-collection gap—it reflects the current ceiling of publicly accessible paired perception–imagery data after principled filtering. A frozen evaluation surface at this scale serves three functions that waiting for more data would not: (1) it prevents architectural conclusions from being drawn in the absence of any benchmark; (2) it creates a fixed comparison point so that gains from future data acquisition can be separated from gains from architectural changes; and (3) it documents the claim boundary explicitly, rather than leaving it implicit or unknown. The comparison ladder and claim-boundary methodology are designed to retain meaning when larger paired data eventually become available.

8 Conclusion

This paper freezes an honest benchmark state for paired fMRI perception–imagery decoding under severe overlap scarcity. On the current max-available paired dataset, Ridge is the strongest overall baseline, shared-only is the strongest canonical neural decoder, and shared-private variants remain exploratory.

The contribution is evaluative and methodological rather than architectural. We provide a reproducible benchmark ladder, expose the current claim boundary, and demonstrate that the present paired regime supports a disciplined qualified interpretation rather than a stronger disentanglement claim.

For the community, the benchmark converts an unstructured data-scarcity problem into three reusable assets: (1) a frozen comparison surface with pre-declared roles, a fixed target space, and a stable evaluation protocol; (2) a claim boundary documenting which conclusions the current overlap regime does and does not support; and (3) a clear recipe for the decisive next experiment—preserve the ladder, acquire materially larger paired data, and determine whether the model ordering changes. Without this benchmark, paired perception–imagery decoding would continue without a fixed reference point, and improvements in future work would be impossible to attribute cleanly to architecture versus evaluation conditions.

Our epistemic position is intentional. We do not claim that disentanglement is wrong in principle. We claim that the current evaluation surface is too scarce to support it. That distinction—between an architectural direction being unjustified *on available evidence* versus being wrong in general—is precisely the kind of claim boundary that an Evaluations and Datasets paper should make explicit.

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A Reproducibility Contract

The benchmark is tied to checked-in canonical configs and workflow entry points. All commands assume execution from the repository root with the project virtual environment active. Config paths below are relative to `configs/canonical/`.

Data preparation.

Acquisition.

```
python -m fmri2img.workflows.acquire_public_nsd_imagery --subjects all
--skip-stimuli --output cache/nsd_imagery_full_all
```

Output: `cache/nsd_imagery_full_all/`

Overlap assembly.

Config: `max_available_overlap.yaml`

```
python -m fmri2img.workflows.prepare_overlap_bootstrap --config ...
--overwrite-existing
```

Output: `outputs/canonical/prepared/full_imagery_overlap/`

Target cache.

Config: `max_available_overlap.yaml`

```
python -m fmri2img.workflows.prepare_targets --config ...
```

Output: `outputs/targets/full_imagery_overlap_vit_l14_image_768.parquet`

Preflight.

Config: `max_available_overlap.yaml`

```
python -m fmri2img.workflows.preflight_data --config ...
```

Output: `outputs/canonical/prepared/full_imagery_overlap/preflight.json`

Baseline and model training.

Ridge baseline.
Config: `max_available_overlap.yaml`
`python -m fmri2img.workflows.run_legacy_ridge_baseline --config ...`
Output: `outputs/canonical/baselines/full_imagery_overlap_ridge_legacy/metrics.json`

Shared-only (Animus Core Decoder).
Config: `animus_core_decoder.yaml`
`python -m fmri2img.workflows.train_animus_core_decoder`
Output: `outputs/animus/core_decoder/train/full_imagery_overlap_shared_only/best_decoder.pt`
Note: Equivalently, shared-only can be trained via the general entry point with `full_imagery_overlap_shared_only.yaml`:
`python -m fmri2img.workflows.train_decoder --config ...`
The seed-stability check (Appendix D) uses this path.

Shared-private, $d_{\text{priv}} = 16$.
Config: `threshold_shared_private_p16.yaml`
`python -m fmri2img.workflows.train_decoder --config ...`
Output: `outputs/research/threshold_shared_private_p16/train/full_imagery_overlap/best_decoder.pt`

Shared-private default.
Config: `max_available_overlap.yaml`
`python -m fmri2img.workflows.train_decoder --config ...`
Output: `outputs/canonical/train/full_imagery_overlap/best_decoder.pt`

Shared-private, $d_{\text{priv}} = 8$.
Config: `max_available_overlap.yaml`
`python -m fmri2img.workflows.train_decoder --config ... --override model.private_dim=8`
Output: `outputs/canonical/train/full_imagery_overlap_priv8/best_decoder.pt`

Shared-private, no domain head (diagnostic).
Config: `max_available_overlap.yaml`
`python -m fmri2img.workflows.train_decoder --config ... --override model.use_domain_head=false`
Output: `outputs/canonical/train/full_imagery_overlap_nodomain/best_decoder.pt`

B Supplementary Material and Artifact Package

An anonymous supplementary package accompanies this submission. It contains:

- The exact YAML configs for every ladder rung (Ridge, shared-only, shared-private $d_{\text{priv}} = 16$).
- A machine-readable artifact manifest (`ARTIFACT_MANIFEST.json`) that records every frozen metric, hyperparameter, external asset citation, and expected output path reported in this paper.
- A step-by-step reproduction README with exact `python -m` commands, environment setup, and required environment variables.
- An external-asset inventory listing every public dataset and pretrained model the benchmark depends on, together with their licenses.
- Seed-stability reproduction instructions for running additional seeds on the two most important neural models (shared-only and SP, $d_{\text{priv}} = 16$).

The supplementary package does not bundle raw NSD data ($\sim$1 TB), derived intermediate datasets, or model checkpoints, because all three are reproducible from the provided configs, commands, and publicly available source data. Full source code will be released upon acceptance.

What a reviewer can verify. From the supplementary material alone, a reviewer can confirm that: (1) every hyperparameter referenced in the paper is present in a checked-in config; (2) every workflow stage has a named entry point and a specific output path; (3) all external data sources are publicly accessible and their licenses are documented; and (4) the evaluation protocol (target space, metrics, ROI contract, model roles) is fully specified in configs rather than in ad-hoc code.

C Additional Notes for Reviewers

Subject breakdown. The 94-row frozen benchmark consists of 38 rows from `subj02`, 18 from `subj03`, 19 from `subj05`, and 19 from `subj07`. The shared paired ID counts by subject are 2, 1, 1, and 1 respectively.

Seed stability. The frozen results reported in Table 1 are from single runs with `seed=0`. Appendix D reports a supplementary seed-stability check for the two primary neural models across three global random seeds. The ordering (shared-only > SP, $d_{\mathrm{priv}} = 16$) is consistent across all seeds, but three seeds do not constitute a formal significance test. The supplementary material includes exact configs and commands for further reproduction.

Broader impacts. The positive value of this work is methodological: it encourages reproducible, under-claimed evaluation in a brain-decoding area where strong narratives can easily outrun the data. The main potential negative impact is that improved brain-decoding workflows can be misinterpreted as supporting stronger privacy or mind-reading capabilities than the evidence justifies. We mitigate that risk by emphasizing benchmark limits, by avoiding subjective-state claims, and by centering claim boundaries rather than deployment claims.

Compute note. The canonical workflows support device-safe execution with automatic fallback to CPU when a requested GPU is unavailable. Per-run wall-clock time on the 94-row benchmark is under 5 minutes for neural models on a single consumer GPU and under 30 seconds for Ridge on CPU. Memory requirements are modest (<4 GB GPU, <8 GB system RAM).

D Seed-Stability Check

To assess whether the benchmark ordering is sensitive to random initialization, we re-ran the two primary neural models—shared-only and SP $d_{\mathrm{priv}} = 16$—with three global random seeds (0, 42, 123). These re-runs use an updated training pipeline that calls `torch_seed_all` before model construction, so weight initialization, data shuffling, and batch sampling are all deterministic per seed. Hardware: NVIDIA H100 80GB; all other hyperparameters are identical to the frozen benchmark.

Table 3: Seed-stability check for the two primary neural models on the frozen overlap benchmark (94 rows, 4 subjects, 5 epochs per run). The ordering shared-only > SP $d_{\mathrm{priv}} = 16$ is consistent across all seeds. Even the worst shared-only seed (0.073) exceeds the best SP $d_{\mathrm{priv}} = 16$ seed (0.048).

Model	Seed	Cosine ↑	MSE ↓	Val loss
Shared-only	0	0.079	0.00240	1.147
Shared-only	42	0.073	0.00241	1.156
Shared-only	123	0.150	0.00221	1.101
Mean ± std		*0.101 ± 0.035*	*0.00234 ± 0.00009*	
SP, $d_{\mathrm{priv}} = 16$	0	0.002	0.00260	25.260
SP, $d_{\mathrm{priv}} = 16$	42	0.048	0.00248	25.229
SP, $d_{\mathrm{priv}} = 16$	123	0.010	0.00258	25.947
Mean ± std		*0.020 ± 0.020*	*0.00255 ± 0.00005*	

Interpretation. The ordering is stable: shared-only outperforms SP $d_{\mathrm{priv}} = 16$ under every seed. The absolute performance varies—shared-only cosine ranges from 0.073 to 0.150—which is expected on a 94-row benchmark and illustrates why this regime requires cautious interpretation rather than headline claims. The gap between model families (shared-only mean 0.101 versus SP $d_{\mathrm{priv}} = 16$

mean 0.020) is larger than the within-family standard deviation for both models, providing evidence that the ordering is not a single-seed artifact. Three seeds are not a significance test, but they move the benchmark from a frozen single-run snapshot toward a reproducibly stable ordering.

Note on absolute values. The seed-stability runs use explicit global seeding (affecting model weight initialization), which was not present in the original frozen runs reported in Table 1. Consequently, absolute metrics differ between the frozen benchmark and this appendix. The relevant finding is the stability of the *ordering*, not the exact values. The supplementary material includes the machine-readable summary (`seed_stability_summary.json`) and all per-seed training histories.

E Bootstrap Confidence Intervals

To assess whether the benchmark ordering is robust to test-set sampling, we compute bootstrap confidence intervals over the 19 held-out test rows. For each of the six seed-stability models (three shared-only seeds, three SP $d_{\text{priv}} = 16$ seeds), we collect per-trial cosine similarities on the test set. We then average across seeds to obtain a seed-averaged per-trial cosine vector for each model family, and bootstrap 10,000 resamples to estimate 95% CIs on the mean cosine and on pairwise differences. Ridge per-trial cosines are taken from the frozen baseline evaluation.

Table 4: Bootstrap 95% confidence intervals on mean test cosine similarity. Seed-averaged values aggregate the three seed runs (0, 42, 123) per model family. All pairwise ordering differences exclude zero.

Model	Mean cosine	95% CI
Ridge	0.552	[0.503, 0.597]
Shared-only (seed-avg.)	0.101	[0.093, 0.111]
SP, $d_{\text{priv}} = 16$ (seed-avg.)	0.020	[0.012, 0.030]

Table 5: Bootstrap 95% CIs on pairwise cosine differences. "Excl. zero" indicates whether the CI excludes zero, i.e., whether the ordering is statistically distinguishable at the 95% level on this test set.

Comparison	Mean diff	95% CI	Excl. zero?
Ridge − Shared-only	+0.451	[0.404, 0.495]	Yes
Shared-only − SP d_{16}	+0.081	[0.074, 0.087]	Yes
Ridge − SP d_{16}	+0.532	[0.487, 0.576]	Yes

Per-seed ordering stability. The pairwise CI between shared-only and SP $d_{\text{priv}} = 16$ excludes zero for every individual seed (seed 0: +0.077, CI [0.063, 0.094]; seed 42: +0.026, CI [0.020, 0.030]; seed 123: +0.140, CI [0.127, 0.151]). Even the weakest single-seed difference (seed 42, +0.026) has a CI that does not include zero, indicating that the ordering is not an artifact of any particular random initialization.

Interpretation. On 19 test rows, bootstrap CIs have limited precision. The intervals reported here should be read as evidence that the ordering is consistent with a real population-level difference rather than as definitive inference. Together with the seed-stability check (Appendix D), the bootstrap analysis provides two independent lines of evidence for the same ordering: Ridge > shared-only > SP $d_{\text{priv}} = 16$. Neither line alone constitutes a formal hypothesis test, but their convergence raises the bar for an alternative explanation.

Per-condition breakdown. The test set contains 3 perception and 16 imagery trials, all from `subj02`. Table 6 reports the per-condition cosine breakdown. Ridge performs similarly on both conditions (perception 0.554, imagery 0.552). Neural models show noisier condition splits, as expected given the 3-trial perception subsample, but the overall ordering is preserved within both conditions.

Table 6: Per-condition test cosine similarity (seed-averaged ± std for neural models). The test set has 3 perception and 16 imagery trials from `subj02`. The small perception count makes per-condition estimates noisy.

Model	Overall	Perception ($n = 3$)	Imagery ($n = 16$)
Ridge	0.552	0.554	0.552
Shared-only	0.101 ± 0.035	0.108 ± 0.026	0.100 ± 0.037
SP, d_{16}	0.020 ± 0.020	0.050 ± 0.006	0.014 ± 0.024

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Section 1 explicitly limit the paper's scope to a frozen benchmark ordering, a claim-boundary methodology, and a reproducibility contribution. No architectural novelty, disentanglement benefit, or statistical significance is claimed.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 7 provides a structured discussion of five threats to validity (overlap scarcity, held-out support weakness, lack of statistical breadth, dependence on recoverable artifacts, and absence of confirmatory threshold evidence) and explicitly argues why they are compatible with the paper's E&D contribution.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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4. **Experimental result reproducibility**

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   Answer: [Yes]

   Justification: Sections 3–5 specify the data regime (94 rows, 4 subjects, 5 paired IDs), ROI groups, target space, hyperparameters, metrics, and evaluation protocol. Appendix A provides exact canonical configs, workflow commands, and artifact paths for every benchmark rung.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: Anonymous supplementary material includes the exact YAML configs for every ladder rung, a machine-readable artifact manifest, step-by-step reproduction instructions, and an external-asset inventory. All source data is publicly available (NSD, NSD-Imagery, COCO, CLIP ViT-L/14). Full source code will be released upon acceptance. See Appendix B.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 4 specifies model roles, the complete hyperparameter family (batch size, epochs, learning rate, weight decay, seed), primary and secondary metrics, and evaluation constraints. Appendix A provides the exact canonical configs.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: Appendix D reports a 3-seed stability check for the two primary neural models. Appendix E provides 10,000-resample bootstrap 95% confidence intervals over the 19-row test set: all three pairwise ordering differences (Ridge–shared-only, shared-only–SP $d_{\text{priv}} = 16$, Ridge–SP $d_{\text{priv}} = 16$) exclude zero. The per-seed ordering CIs also exclude zero for every individual seed. We do not claim formal significance testing beyond the bootstrap; together, seed stability and bootstrap CIs provide two independent lines of evidence for the ordering.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: Appendix C reports approximate per-run wall-clock time ($<$5 min neural on consumer GPU, $<$30 s Ridge on CPU) and memory requirements ($<$4 GB GPU, $<$8 GB system RAM). Workflows support automatic CPU fallback.

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   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

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    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: Appendix C discusses the methodological value of disciplined benchmark reporting and the risk of overstating brain-decoding capabilities.

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    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

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Justification: All external assets are cited in the paper. The supplementary material (Appendix B) includes a complete external-asset inventory listing each dataset and pretrained model with its source URL and license (NSD: CC BY 4.0/NSD terms; COCO: CC BY 4.0; CLIP ViT-L/14: MIT).

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

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